CLASS 8 — MATHS · FORMULA SHEET

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Quick revision, one page

Baudhayana-Pythagoras theorem

- In a right-angled triangle: (hypotenuse)^2 = (base)^2+ (perpendicular)^2, i.e. $c^2 = a^2 + b^2$.
- The hypotenuse is the side opposite the right angle and is the longest side.

Converse

- If $c^2 = a^2 + b^2$ for the sides of a triangle with c the longest, the triangle is right-angled and the right angle is opposite c.

Pythagorean triplets

- Common triplets: $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, $(20, 21, 29)$; multiples of a triplet are triplets.
- For $m > 1$: $(2m,\ m^2 - 1,\ m^2 + 1)$ is a Pythagorean triplet.

Applications

- Diagonal of a rectangle $= \sqrt{l^2 + b^2}$; diagonal of a square of side $a = a\sqrt{2}$.
- Rhombus with diagonals d_1, d_2: side $= \sqrt{\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2}$.
- Ladder/pole/wall problems: form a right triangle with ground, wall, ladder as the hypotenuse.

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